

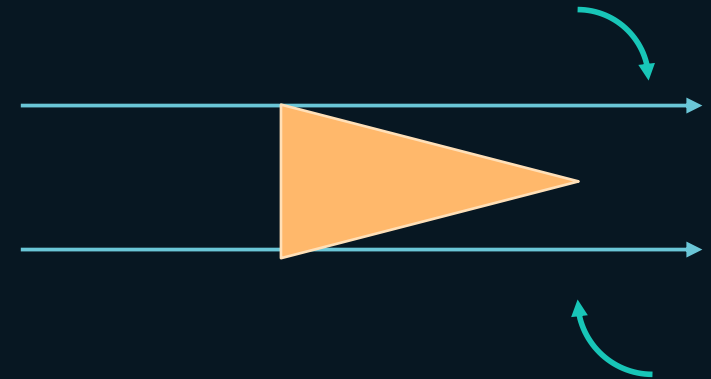
Numerical Optimization of Bluff-Body Geometry

for Controlled Vortex Shedding in a Vortex Flowmeter

Extension-based Numerical Methods Project

Group Members: [Name 1], [Name 2], [Name 3], [Name 4], [Name 5]

Section/Subsection: [e.g., A1 / A2]



Flow → bluff body → alternating vortices

Base Paper

Shape Optimization of the Triangular Vortex Flowmeter Based on the LBM Method

Sun et al., Symmetry, 2025

Domain

Fluid dynamics / vortex flowmeter design

Core Method

Lattice Boltzmann Method for simulating flow and vortex shedding

Paper Goal

Optimize triangular bluff-body shape for stronger vortex shedding and lower pressure loss

Why this paper fits our project

It provides a recent CFD/simulation base and leaves room for a simpler course-level numerical optimization extension.

Problem Statement

A vortex flowmeter needs a bluff body that creates regular vortex shedding, but excessive pressure loss reduces efficiency.

Project Scope

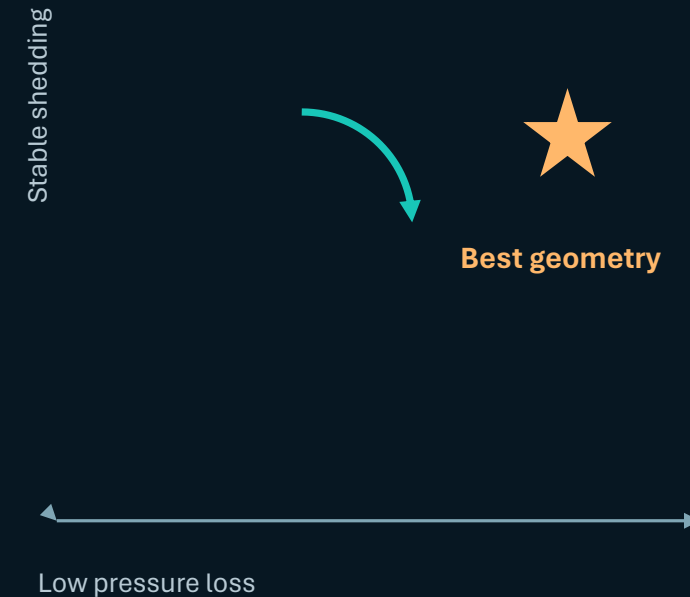
We will perform an extension-based study using a simplified 2D simulation and numerical optimization instead of reproducing the full high-fidelity LBM setup.

Extension Type

Methodological Extension

Geometry → Simulation → Vortex Metrics → Optimization

Target trade-off



Expected Methodology



Final output

A comparison of baseline vs optimized bluff-body geometry using vortex stability, shedding frequency, pressure loss, and convergence/error plots.

Numerical methods used: simulation, interpolation, optimization, data fitting, and error analysis.